

PROJECT TITLE

AI-Based Fake News Detection System

1. INTRODUCTION

Fake news has become a major issue in the digital age. With the rapid growth of social media and online platforms, misleading and false information spreads quickly and influences public opinion. This creates confusion and can lead to incorrect decisions.

To address this problem, an AI-based system is developed to automatically classify news as *Real* or *Fake* using machine learning techniques.

2. PROBLEM STATEMENT

There is no easy and reliable way for users to verify whether a news article is genuine or fake. Manual verification is time-consuming and often inaccurate. Therefore, an automated system is required to analyze news content and determine its authenticity.

3. OBJECTIVE

- To classify news articles as Fake or Real
- To apply machine learning techniques on textual data
- To provide quick and accurate predictions to users

4. METHODOLOGY

- A dataset containing fake and real news is used
- Text preprocessing is performed:
 - Converting text to lowercase
 - Removing stopwords

- Feature extraction is done using **TF-IDF Vectorization**
- A machine learning model (**Logistic Regression**) is trained
- The trained model predicts the category of new input text

5. IMPLEMENTATION

The system is implemented using Python and Scikit-learn.

Steps followed:

1. Load the dataset
2. Clean and preprocess text data
3. Apply TF-IDF vectorization
4. Train the machine learning model
5. Take user input and generate prediction

6. CHALLENGES

- Cleaning unstructured text data
- Improving model accuracy
- Handling different writing styles of news

7. RESULT

The model successfully classifies news as fake or real with reasonable accuracy.

It provides instant predictions based on user input.

8. CONCLUSION

This project demonstrates how machine learning can be used to detect fake news effectively. It helps users verify information quickly and reduces the spread of misinformation.

9. FUTURE SCOPE

- Use advanced deep learning models
- Support multiple languages
- Integrate with real-time news APIs